

# Electromagnetism Test Document for VimTeX, SyncTeX, LSP, and Snippets

Generated Test File

June 28, 2026

## Abstract

This document is a deliberately equation-heavy electromagnetism sample for testing a LaTeX editing setup. It includes cross-references, citations, displayed equations, aligned environments, a bibliography, TikZ figures, tables, labels, and enough page breaks to make forward and inverse SyncTeX testing meaningful. Compile with `latexmk -pdf -synctex=1 em-test.tex`. For VimTeX, try `\ll` to compile, `\lv` to view, `\le` for errors, and PDF-to-source inverse search in your viewer.

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# 1 Electrostatics and Coulomb's Law

Electrostatics begins with the empirical observation that stationary charges exert forces on one another. For point charges  $q_1$  and  $q_2$  separated by the vector  $\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2$ , Coulomb's law is

$$\mathbf{F}_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{\|\mathbf{r}\|^2} \hat{\mathbf{r}}. \quad (1)$$

This is useful for force calculations, but field language is usually cleaner. The electric field due to a point charge  $q$  at the origin is

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}. \quad (2)$$

For a continuous charge distribution  $\rho(\mathbf{r}')$ , the superposition principle gives

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int_{\mathbb{R}^3} \rho(\mathbf{r}') \frac{\mathbf{r} - \mathbf{r}'}{\|\mathbf{r} - \mathbf{r}'\|^3} d^3 r'. \quad (3)$$

A useful test for editor navigation is to jump between [equation \(1\)](#), [equation \(2\)](#), and [equation \(3\)](#). The conceptual move from forces to fields is emphasized in standard treatments such as [\[1\]](#), [\[2\]](#).

## 1.1 Potential energy and electric potential

Because the electrostatic field is conservative, it may be written as the negative gradient of a scalar potential:

$$\mathbf{E} = -\nabla V. \quad (4)$$

The potential generated by a charge distribution is

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}')}{\|\mathbf{r} - \mathbf{r}'\|} d^3 r'. \quad (5)$$

This expression is often easier to evaluate than the vector integral in [equation \(3\)](#), after which the field can be recovered from [equation \(4\)](#).

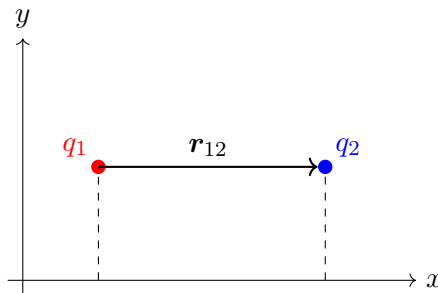


Figure 1: Two point charges separated by a displacement vector. Use this figure to test SyncTeX jumps near captions, equations, and text.

See [figure 1](#) and test whether clicking near the caption in the PDF jumps to this source location.

## 2 Gauss's Law

Gauss's law is a compact expression of the relationship between electric flux and enclosed charge. In integral form,

$$\oint_{\partial\Omega} \mathbf{E} \cdot d\mathbf{a} = \frac{Q_{\text{enc}}}{\varepsilon_0}. \quad (6)$$

Using the divergence theorem,

$$\oint_{\partial\Omega} \mathbf{E} \cdot d\mathbf{a} = \int_{\Omega} \nabla \cdot \mathbf{E} \, d^3r, \quad (7)$$

and since  $Q_{\text{enc}} = \int_{\Omega} \rho \, d^3r$ , one obtains the differential form

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}. \quad (8)$$

### 2.1 Spherical symmetry

For a point charge or uniformly charged sphere, symmetry forces the field to be radial:

$$\mathbf{E}(\mathbf{r}) = E(r)\hat{\mathbf{r}}. \quad (9)$$

A spherical Gaussian surface gives

$$\oint \mathbf{E} \cdot d\mathbf{a} = E(r)4\pi r^2, \quad (10)$$

$$E(r)4\pi r^2 = \frac{q}{\varepsilon_0}, \quad (11)$$

$$E(r) = \frac{1}{4\pi\varepsilon_0} \frac{q}{r^2}. \quad (12)$$

This recovers [equation \(2\)](#), but with far less calculation.

### 2.2 Planar and cylindrical symmetry

For an infinite charged sheet with surface charge density  $\sigma$ , a pillbox Gaussian surface gives

$$E = \frac{\sigma}{2\varepsilon_0}. \quad (13)$$

For an infinite line charge with linear density  $\lambda$ , a cylindrical Gaussian surface gives

$$E(s) = \frac{\lambda}{2\pi\varepsilon_0 s}. \quad (14)$$

These examples are good places to test text-object motions, surrounding delimiters, and snippets for equations.

Table 1: Common Gaussian surfaces and resulting electric fields.

| Source                  | Gaussian surface | Field magnitude                 |
|-------------------------|------------------|---------------------------------|
| Point charge $q$        | sphere           | $q/(4\pi\varepsilon_0 r^2)$     |
| Infinite line $\lambda$ | cylinder         | $\lambda/(2\pi\varepsilon_0 s)$ |
| Infinite sheet $\sigma$ | pillbox          | $\sigma/(2\varepsilon_0)$       |

[Table 1](#) is included to test table navigation, captions, and references.

### 3 Boundary Conditions and Conductors

In electrostatic equilibrium, the electric field inside a conductor is zero:

$$\mathbf{E}_{\text{inside}} = \mathbf{0}. \quad (15)$$

If this were not true, free charges would accelerate until they rearranged themselves to cancel the internal field.

The tangential component of the electrostatic field is continuous across a boundary:

$$\hat{\mathbf{n}} \times (\mathbf{E}_2 - \mathbf{E}_1) = \mathbf{0}. \quad (16)$$

The normal component jumps by an amount determined by the surface charge density:

$$\hat{\mathbf{n}} \cdot (\mathbf{E}_2 - \mathbf{E}_1) = \frac{\sigma}{\varepsilon_0}. \quad (17)$$

Together, [equations \(16\)](#) and [\(17\)](#) are among the most useful local constraints in electrostatics.

#### 3.1 Capacitance

For two conductors carrying charges  $+Q$  and  $-Q$ , the capacitance is

$$C = \frac{Q}{\Delta V}. \quad (18)$$

For a parallel-plate capacitor with plate area  $A$  and separation  $d$ , neglecting fringing fields,

$$C = \frac{\varepsilon_0 A}{d}. \quad (19)$$

The stored electrostatic energy can be written as

$$U = \frac{1}{2}CV^2 = \frac{Q^2}{2C} = \frac{1}{2}QV. \quad (20)$$

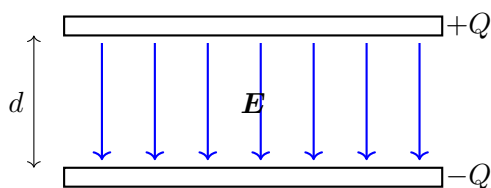


Figure 2: Idealized parallel-plate capacitor. The field is approximately uniform between the plates.

Try jumping from [equation \(19\)](#) to [figure 2](#) using your LSP or Telescope symbol tools.

## 4 Poisson and Laplace Equations

Combining  $\mathbf{E} = -\nabla V$  with Gauss's law yields Poisson's equation:

$$\nabla^2 V = -\frac{\rho}{\varepsilon_0}. \quad (21)$$

In regions with no charge density, this reduces to Laplace's equation:

$$\nabla^2 V = 0. \quad (22)$$

### 4.1 Uniqueness

A central result is that solutions to Laplace's equation are unique once the boundary values are specified. Suppose  $V_1$  and  $V_2$  both solve Laplace's equation in a region  $\Omega$  and agree on  $\partial\Omega$ . Define  $U = V_1 - V_2$ . Then

$$\nabla^2 U = 0, \quad U|_{\partial\Omega} = 0. \quad (23)$$

Using Green's first identity,

$$\int_{\Omega} \|\nabla U\|^2 d^3r = \oint_{\partial\Omega} U \nabla U \cdot d\mathbf{a} - \int_{\Omega} U \nabla^2 U d^3r = 0. \quad (24)$$

Therefore  $\nabla U = \mathbf{0}$ , and the boundary condition forces  $U = 0$ , so  $V_1 = V_2$ .

### 4.2 Separation of variables

In Cartesian coordinates,

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}. \quad (25)$$

For a rectangular region, one may try

$$V(x, y, z) = X(x)Y(y)Z(z), \quad (26)$$

which turns the partial differential equation into ordinary differential equations. This section is useful for testing snippets involving partial derivatives and aligned equations.

$$\frac{1}{X} \frac{d^2 X}{dx^2} + \frac{1}{Y} \frac{d^2 Y}{dy^2} + \frac{1}{Z} \frac{d^2 Z}{dz^2} = 0. \quad (27)$$

## 5 Magnetostatics

Magnetostatics describes steady currents and time-independent magnetic fields. The Lorentz force on a charge  $q$  moving with velocity  $\mathbf{v}$  is

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \quad (28)$$

For a current distribution  $\mathbf{J}(\mathbf{r}')$ , the Biot–Savart law gives

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \mathbf{J}(\mathbf{r}') \times \frac{\mathbf{r} - \mathbf{r}'}{\|\mathbf{r} - \mathbf{r}'\|^3} d^3r'. \quad (29)$$

### 5.1 Ampere’s law

For steady currents, Ampere’s law is

$$\oint_{\partial S} \mathbf{B} \cdot d\boldsymbol{\ell} = \mu_0 I_{\text{enc}}. \quad (30)$$

Using Stokes’ theorem gives

$$\boldsymbol{\nabla} \times \mathbf{B} = \mu_0 \mathbf{J}. \quad (31)$$

The magnetic field of a long straight wire follows immediately:

$$B(s) = \frac{\mu_0 I}{2\pi s}. \quad (32)$$

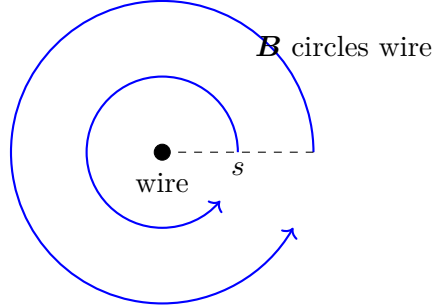


Figure 3: Magnetic field lines around a long straight current-carrying wire.

Magnetostatic calculations often mirror electrostatic ones, but with curls replacing divergences. Compare [equation \(8\)](#) with [equation \(31\)](#).

## 6 Vector Potential and Gauge Freedom

Because magnetic fields are solenoidal,

$$\nabla \cdot \mathbf{B} = 0, \quad (33)$$

one may define a vector potential  $\mathbf{A}$  such that

$$\mathbf{B} = \nabla \times \mathbf{A}. \quad (34)$$

The vector potential is not unique. If  $\chi$  is a scalar function, then

$$\mathbf{A}' = \mathbf{A} + \nabla \chi \quad (35)$$

produces the same magnetic field because

$$\nabla \times \mathbf{A}' = \nabla \times \mathbf{A} + \nabla \times \nabla \chi = \nabla \times \mathbf{A}. \quad (36)$$

### 6.1 Coulomb gauge

One common choice is the Coulomb gauge,

$$\nabla \cdot \mathbf{A} = 0. \quad (37)$$

In magnetostatics, combining [equations \(31\)](#), [\(34\)](#) and [\(37\)](#) yields

$$\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}, \quad (38)$$

$$-\nabla^2 \mathbf{A} = \mu_0 \mathbf{J}, \quad (39)$$

$$\nabla^2 \mathbf{A} = -\mu_0 \mathbf{J}. \quad (40)$$

This resembles Poisson's equation in [equation \(21\)](#), but applies component-wise to the vector potential.

### 6.2 Aharonov–Bohm intuition

Even where  $\mathbf{B} = \mathbf{0}$ , the vector potential may influence quantum phases. This is one reason the vector potential is more than a mere computational convenience in quantum mechanics. A classical electromagnetism course may introduce this only briefly, while more advanced treatments discuss it in detail [\[3\]](#).



## 7 Time-Dependent Fields and Maxwell's Correction

The magnetostatic form of Ampere's law is incomplete for time-dependent situations. Taking the divergence of  $\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$  gives

$$0 = \mu_0 \nabla \cdot \mathbf{J}, \quad (41)$$

which would require charge conservation in the restricted form  $\nabla \cdot \mathbf{J} = 0$ . But the continuity equation is

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0. \quad (42)$$

Using Gauss's law,

$$\rho = \varepsilon_0 \nabla \cdot \mathbf{E}, \quad (43)$$

the continuity equation becomes

$$\nabla \cdot \left( \mathbf{J} + \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) = 0. \quad (44)$$

Maxwell's corrected Ampere law is therefore

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}. \quad (45)$$

### 7.1 Faraday's law

Faraday's law relates changing magnetic flux to induced electric fields:

$$\oint_{\partial S} \mathbf{E} \cdot d\boldsymbol{\ell} = - \frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{a}. \quad (46)$$

In differential form,

$$\nabla \times \mathbf{E} = - \frac{\partial \mathbf{B}}{\partial t}. \quad (47)$$

This is a good place to test diagnostics by intentionally deleting a brace, then using `\le` to inspect errors.

## 8 Maxwell's Equations

In vacuum with sources, Maxwell's equations are

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}, \quad (48)$$

$$\nabla \cdot \mathbf{B} = 0, \quad (49)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad (50)$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}. \quad (51)$$

Together these equations unify electrostatics, magnetostatics, induction, and electromagnetic waves. The source-free case is obtained by setting  $\rho = 0$  and  $\mathbf{J} = \mathbf{0}$ .

### 8.1 Constitutive relations

In matter, it is often more useful to introduce  $\mathbf{D}$  and  $\mathbf{H}$ :

$$\mathbf{D} = \varepsilon_0 \mathbf{E} + \mathbf{P}, \quad (52)$$

$$\mathbf{H} = \frac{1}{\mu_0} \mathbf{B} - \mathbf{M}. \quad (53)$$

For simple linear media,

$$\mathbf{D} = \varepsilon \mathbf{E}, \quad (54)$$

$$\mathbf{B} = \mu \mathbf{H}. \quad (55)$$

Then the source equations become

$$\nabla \cdot \mathbf{D} = \rho_{\text{free}}, \quad (56)$$

$$\nabla \times \mathbf{H} = \mathbf{J}_{\text{free}} + \frac{\partial \mathbf{D}}{\partial t}. \quad (57)$$

Table 2: Differential Maxwell equations in vacuum.

| Equation                                                                                           | Meaning                                                      |
|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$                                                     | electric charge sources electric field                       |
| $\nabla \cdot \mathbf{B} = 0$                                                                      | no magnetic monopoles in classical EM                        |
| $\nabla \times \mathbf{E} = -\partial \mathbf{B}/\partial t$                                       | changing magnetic field induces electric circulation         |
| $\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \partial \mathbf{E}/\partial t$ | current and displacement current source magnetic circulation |

Use [table 2](#) to test citation, table, and reference completion.

## 9 Electromagnetic Waves

In vacuum with no charges or currents, Maxwell's equations imply wave equations for both fields. Taking the curl of Faraday's law gives

$$\nabla \times (\nabla \times \mathbf{E}) = -\frac{\partial}{\partial t}(\nabla \times \mathbf{B}), \quad (58)$$

$$\nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} = -\mu_0 \varepsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}. \quad (59)$$

Since  $\nabla \cdot \mathbf{E} = 0$ , this becomes

$$\nabla^2 \mathbf{E} - \mu_0 \varepsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} = 0. \quad (60)$$

Similarly,

$$\nabla^2 \mathbf{B} - \mu_0 \varepsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2} = 0. \quad (61)$$

The wave speed is

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}. \quad (62)$$

### 9.1 Plane waves

A monochromatic plane wave propagating in the  $+\hat{z}$  direction may be written

$$\mathbf{E}(z, t) = E_0 \cos(kz - \omega t) \hat{\mathbf{x}}, \quad (63)$$

$$\mathbf{B}(z, t) = \frac{E_0}{c} \cos(kz - \omega t) \hat{\mathbf{y}}. \quad (64)$$

The fields are transverse, mutually perpendicular, and related by

$$\mathbf{B} = \frac{1}{c} \hat{\mathbf{k}} \times \mathbf{E}. \quad (65)$$

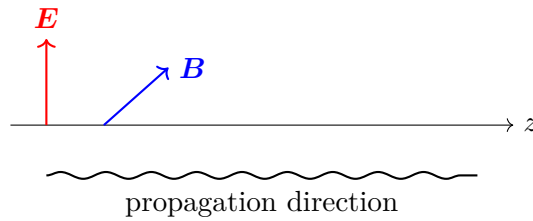


Figure 4: Schematic transverse electromagnetic plane wave.

The derivation of [equations \(60\)](#) and [\(61\)](#) is a useful stress test for equation alignment, references, and SyncTeX movement.

## 10 Energy, Momentum, and the Poynting Vector

The electromagnetic energy density in vacuum is

$$u = \frac{1}{2}\varepsilon_0 E^2 + \frac{1}{2\mu_0} B^2. \quad (66)$$

The Poynting vector is

$$\mathbf{S} = \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B}. \quad (67)$$

It represents electromagnetic energy flux.

### 10.1 Poynting theorem

Starting from Maxwell's equations, one can derive

$$\frac{\partial u}{\partial t} + \nabla \cdot \mathbf{S} = -\mathbf{J} \cdot \mathbf{E}. \quad (68)$$

The term  $\mathbf{J} \cdot \mathbf{E}$  is the rate per unit volume at which fields do work on charges. If  $\mathbf{J} = \mathbf{0}$ , then electromagnetic energy is locally conserved:

$$\frac{\partial u}{\partial t} + \nabla \cdot \mathbf{S} = 0. \quad (69)$$

### 10.2 Radiation pressure

Electromagnetic waves also carry momentum. For a wave normally incident on a perfectly absorbing surface, the radiation pressure is

$$p = \frac{\langle S \rangle}{c}. \quad (70)$$

For a perfectly reflecting surface, the pressure doubles:

$$p = \frac{2\langle S \rangle}{c}. \quad (71)$$

This section is good for testing references to late-document equations such as [equations \(67\)](#) and [\(68\)](#).

## 11 Potentials, Retardation, and a Final Checklist

For time-dependent fields, the scalar and vector potentials are often written

$$\mathbf{E} = -\nabla V - \frac{\partial \mathbf{A}}{\partial t}, \quad (72)$$

$$\mathbf{B} = \nabla \times \mathbf{A}. \quad (73)$$

In the Lorenz gauge,

$$\nabla \cdot \mathbf{A} + \mu_0 \varepsilon_0 \frac{\partial V}{\partial t} = 0, \quad (74)$$

the potentials satisfy wave equations:

$$\nabla^2 V - \mu_0 \varepsilon_0 \frac{\partial^2 V}{\partial t^2} = -\frac{\rho}{\varepsilon_0}, \quad (75)$$

$$\nabla^2 \mathbf{A} - \mu_0 \varepsilon_0 \frac{\partial^2 \mathbf{A}}{\partial t^2} = -\mu_0 \mathbf{J}. \quad (76)$$

Their solutions are retarded potentials:

$$V(\mathbf{r}, t) = \frac{1}{4\pi\varepsilon_0} \int \frac{\rho(\mathbf{r}', t_r)}{\|\mathbf{r} - \mathbf{r}'\|} d^3r', \quad (77)$$

$$\mathbf{A}(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}', t_r)}{\|\mathbf{r} - \mathbf{r}'\|} d^3r', \quad (78)$$

where

$$t_r = t - \frac{\|\mathbf{r} - \mathbf{r}'\|}{c}. \quad (79)$$

### 11.1 Editing setup checklist

Use this document to test the following workflow items:

1. Compile continuously with `\ll`.
2. Open or forward-search the PDF with `\lv`.
3. Jump from PDF back to source using inverse SyncTeX.
4. Inspect LaTeX errors with `\le`.
5. Open VimTeX information with `\li` or `:VimtexInfo`.
6. Search symbols such as `Maxwell` and `Poynting` with Telescope or LSP document symbols.
7. Test snippets inside math zones, for example fractions, hats, bars, bras, kets, and display equations.
8. Test citations such as [\[1\]](#), [\[2\]](#), and [\[3\]](#).

## 11.2 Deliberate navigation targets

Here are some source locations that should be easy to find from the PDF:

- [Equation \(1\)](#): Coulomb’s law.
- [Equation \(8\)](#): Gauss’s law in differential form.
- [Equation \(29\)](#): Biot–Savart law.
- [Equation \(51\)](#): Maxwell–Ampere law.
- [Equation \(60\)](#): electromagnetic wave equation.
- [Equation \(67\)](#): Poynting vector.
- [Equation \(79\)](#): retarded time.

## References

- [1] D. J. Griffiths, *Introduction to Electrodynamics*, 4th ed. Cambridge University Press, 2017.
- [2] E. M. Purcell and D. J. Morin, *Electricity and Magnetism*, 3rd ed. Cambridge University Press, 2013.
- [3] J. D. Jackson, *Classical Electrodynamics*, 3rd ed. Wiley, 1999.